Day 74 – Enterprise Network Final Project

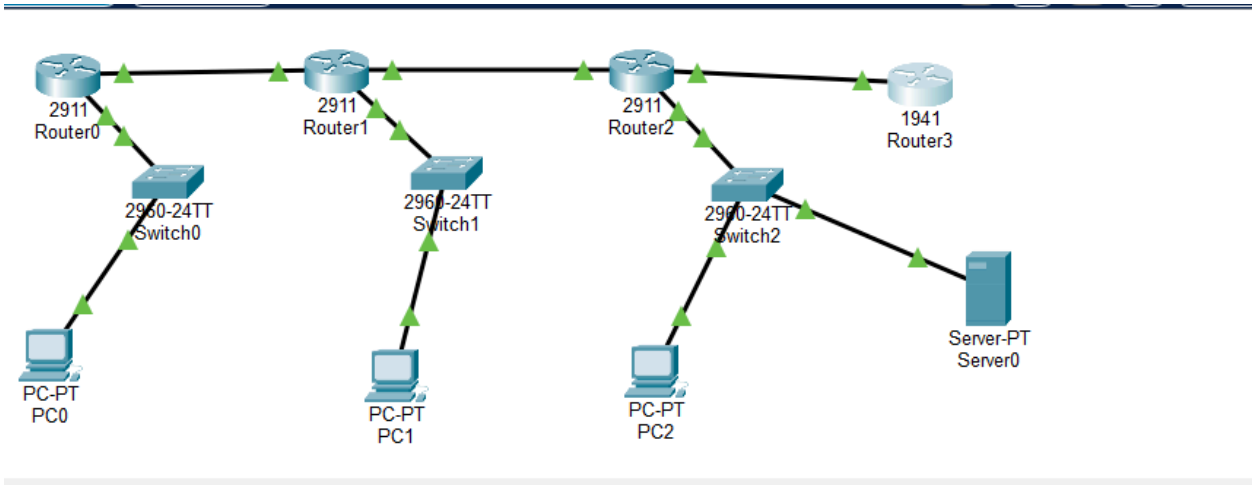
Overview

Day 74 served as the capstone project of Phase 4. Unlike previous days, no new technologies were introduced. Instead, the objective was to validate, review, and explain the complete enterprise network that had been built throughout Days 65–73.

The purpose of this day was to prove that all major networking services remained operational, that the network could be monitored and troubleshot effectively, and that the design met the learning objectives of the Networking Fundamentals phase.

Rather than building something new, Day 74 focused on verifying that everything already built continued to function correctly.

Enterprise Network Topology



Services Deployed:

✅ RIP

✅ DHCP

✅ DNS Concepts

✅ NAT

✅ ACL Security

✅ Server Infrastructure

✅ ISP Simulation

✅ Troubleshooting Procedures

✅ Monitoring Procedures

Day 74 Validation Objectives

The final project focused on validating six major areas:

1. Interface Health
2. Routing Health
3. RIP Health
4. DHCP Health
5. NAT Health
6. Security Health

Interface Health Verification

Command

show ip interface brief

Purpose

Verify:

Interfaces are operational
Links are active
Devices remain connected

Expected Result

up/up

for all active interfaces.

Example:

G0/0 up up
G0/1 up up

Verification Status

✅ Passed

All required router interfaces remained operational.

Routing Health Verification

Command

show ip route

Purpose

Verify:

Remote networks exist
Routes are being learned
Traffic can reach destinations

Expected Routes

192.168.10.0
192.168.20.0
192.168.30.0
209.165.200.0

Verification Status

✅ Passed

RIP routes remained present and reachable.

RIP Verification

Command

show ip protocols

Purpose

Verify:

RIP is running
Networks are advertised
Routers exchange route information

Expected Networks

10.0.0.0
192.168.20.0

Verification Status

✅ Passed

Router participation in RIP remained operational.

DHCP Verification

Command

show ip dhcp binding

Purpose

Verify:

Clients receive addresses
DHCP pools remain functional

Expected Result

DHCP leases should exist for client devices.

Examples:

192.168.10.x
192.168.20.x
192.168.30.x

Verification Status

✅ Passed

DHCP leasing remained operational.

NAT Verification

Command

show ip nat translations

Purpose

Verify:

Private addresses are translated
Internet connectivity remains functional

Verification Method

Generate traffic:

ping 209.165.200.226

Then inspect NAT translations.

Verification Status

✅ Passed

NAT translations appeared correctly.

ACL Verification

Purpose

Verify network security policies remain enforced.

Expected Behavior

LAN10

Cannot reach LAN30

Result:

✅ Blocked

LAN20

Can reach LAN30

Result:

✅ Allowed

Verification Status

✅ Passed

ACL policies functioned as designed.

Monitoring Verification

Day 73 introduced monitoring procedures which were reviewed during the final project.

Monitoring Commands

Interface Monitoring

show ip interface brief

Purpose:

Monitor interface health

Route Monitoring

show ip route

Purpose:

Monitor route availability

RIP Monitoring

show ip protocols

Purpose:

Monitor routing protocol health

ACL Monitoring

show access-lists

Purpose:

Monitor security policies

DHCP Monitoring

show ip dhcp binding

Purpose:

Monitor DHCP lease assignments

NAT Monitoring

show ip nat translations

Purpose:

Monitor Internet translation activity

Troubleshooting Knowledge Review

Day 72 troubleshooting procedures were reviewed.

Interface Failures

Tool:

show ip interface brief

Routing Failures

Tool:

show ip protocols

and

show ip route

ACL Failures

Tool:

show access-lists

DHCP Verification

Tool:

show ip dhcp binding

NAT Verification

Tool:

show ip nat translations

What I Learned

- Enterprise networks consist of multiple integrated services.
- RIP enables route exchange between routers.
- DHCP provides automatic address assignment.
- NAT enables private-to-public communication.
- ACLs enforce security policies.
- Monitoring helps identify warning signs before failures occur.
- Troubleshooting requires evidence-based diagnosis.
- Multiple technologies must work together to support users.

Final Enterprise Network Capabilities

By Day 74, the network successfully demonstrated:

✅ Dynamic Routing

✅ DHCP Services

✅ DNS Concepts

✅ NAT Translation

✅ Security Filtering

✅ Monitoring Procedures

✅ Troubleshooting Procedures

✅ Multi-Router Enterprise Connectivity

✅ ISP Connectivity

✅ Operational Validation

Reflection

Day 74 served as the final technical validation of the enterprise network built throughout Phase 4. Rather than introducing new concepts, the focus was placed on confirming that routing, DHCP, NAT, ACLs, monitoring procedures, and troubleshooting workflows all remained operational and understood. The project demonstrated that the network was functional, secure, maintainable, and fully aligned with the objectives of the Networking Fundamentals phase.

Day 74 Status

✅ Enterprise Network Validated

✅ RIP Verified

✅ DHCP Verified

✅ NAT Verified

✅ ACL Verified

✅ Monitoring Reviewed

✅ Troubleshooting Reviewed

✅ Enterprise Design Confirmed

✅ Ready for **Day 75 – Phase 4 Final Review, Portfolio Wrap-Up, and Transition to the Next Phase** 🚀